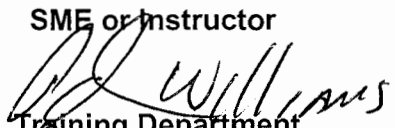
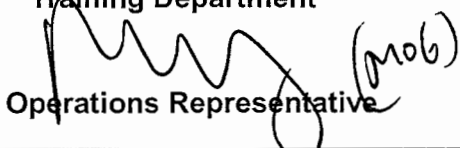


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	Salem Generating Station		
SYSTEM:	Emergency Core Cooling Systems		
TASK:	Raise ECCS Accumulator Level and Pressure		
TASK NUMBER:	0065010101		
JPM NUMBER:	14-01 NRC Sim c		
ALTERNATE PATH:	☐	K/A NUMBER:	006 A4.07
APPLICABILITY:		IMPORTANCE FACTOR:	
EO ☐	RO ☒	STA ☐	SRO ☒
EVALUATION SETTING/METHOD:	Simulator / Perform		
REFERENCES:	S2.OP-SO.SJ-0002, Accumulator Operations, Rev. 24 (Rev checked 9-23-15)		
TOOLS AND EQUIPMENT:	None		
VALIDATED JPM COMPLETION TIME:	<u>13 minutes</u>		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	<u>N/A</u>		
Developed By:	G Gauding Instructor	Date:	8-13-15
Validated By:	S Bickhart / J Militti SME or Instructor	Date:	9-23-15
Approved By:	 Training Department	Date:	10/7/15
Approved By:	 Operations Representative (mob)	Date:	10/9/15
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE: ☐ SAT ☐ UNSAT			
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level and Pressure with a Safety Injection Pump

TASK 0065010101

NUMBER:

SIMULATOR SETUP: IC-243

*****Mark up simulator laminated copy of S2.OP-SO.SJ-0002, Accumulator Operations to N/A all sections except Sections 5.2 (Raise level with 21 SI pump) and 5.10 (Raise pressure)**

INITIAL

CONDITIONS:

1. The plant is at 100% power with all systems in their normal alignment with control systems in automatic.
2. 24 Accumulator is at 58.4% level and 598.8 psig.
3. The Safety Injection System is available.
4. Nitrogen System is available.
5. RWST Boron Concentration is 2455 ppm.
6. Components Off Normal and Off Normal Tagged has been checked SAT.

INITIATING CUE:

Raise 24 ECCS Accumulator parameters to 60% level using 21 SI pump, and raise 24 Accumulator pressure to 625 psig using IAW S2.OP-SO.SJ-0002, Accumulator Operations.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Raise 24 Accumulator level to 60% using a SI pump, and return system to normal lineup.
2. Raise 24 Accumulator pressure to 625 psig after level has been raised.

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
			Reviews procedure, and initials all Precautions and Limitations. Note: Precaution and Limitation 3.1 states that if both level and pressure need adjustment, level should be adjusted first.		
	5.2.1	ENSURE either of the following conditions exist: • ALL RCS Cold Leg Temperatures >312°F <u>OR</u> • The Reactor Vessel Head is Removed.	Determines ALL RCS Cold Leg Temperatures >312°F from control console indications.		
	5.2.2	<u>IF</u> RCS Pressure <2000 psig, <u>THEN</u> ENSURE CLOSED 21SJ134, COLD LEG DISCHARGE.	Determines RCS pressure is >2000 psig from console indication and marks step N/A..		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.3	START 21 Safety Injection Pump.	Operator depresses 21 Safety Injection Pump START PB and verifies red start light illuminates, green stop light extinguishes, and pump amps indicated. Cue if required: IF operator asks for a Field Operator to do pre-start checks on 21 SI pump, <u>THEN</u> report as NEO that 21 SI Pump is ready for start.		
*	5.2.4	OPEN 2SJ53, 21 SI PUMP DISCHARGE TEST LINE VALVE.	Depresses 2SJ53 21 SI PUMP DISCHARGE TEST LINE VALVE OPEN PB, and verifies open light illuminates and shut light extinguishes.		
*	5.2.5	OPEN 2SJ123, TEST LINE TO CVCS HUT.	Depresses and holds 2SJ123 TEST LINE TO CVCS HUT OPEN PB until open light illuminates and shut light extinguishes.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.6	OPEN associated Accumulator fill valve: ♦ 21SJ20, 21 ACCUMULATOR FILL ♦ 22SJ20, 22 ACCUMULATOR FILL ♦ 23SJ20, 23 ACCUMULATOR FILL ♦ 24SJ20, 24 ACCUMULATOR FILL	Depresses 24SJ20 24 ACCUMULATOR FILL OPEN PB and verifies open light illuminates and shut light extinguishes. Monitors level on 2LI935D and 2LI934D. Note: It will take just over a minute to raise 24 Accumulator level to 60%.		
*	5.2.7	When desired level is reached, CLOSE the associated Accumulator fill valve: ♦ 21SJ20, 21 ACCUMULATOR FILL ♦ 22SJ20, 22 ACCUMULATOR FILL ♦ 23SJ20, 23 ACCUMULATOR FILL ♦ 24SJ20, 24 ACCUMULATOR FILL	When 24 Accumulator level indicates 60% on 2LI935D and 2LI934D, depresses 24SJ20 24 ACCUMULATOR FILL CLOSE PB and verifies red open light extinguished and green shut light illuminates.		
	5.2.8	If required, REPEAT Steps 5.2.6 and 5.2.7 to fill additional Accumulators.	Determines no other Accumulators will be filled.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.2.9	CLOSE 2SJ53.	Depresses 2SJ53 CLOSE PB and verifies red open light extinguished and green shut light illuminates.		
*	5.2.10	CLOSE 2SJ123.	Depresses 2SJ123 CLOSE PB and verifies red open light extinguished and green shut light illuminates.		
*	5.2.11	STOP 21 Safety Injection Pump.	Depresses 21 SI Pump STOP PB and verifies green stop light illuminate and red start light extinguishes.		
	5.2.12	<u>IF</u> in Modes 1, 2, or 3 <u>AND</u> 21SJ134 was CLOSED in Step 5.2.2, <u>THEN</u> OPEN 21SJ134.	Determines 21SJ134 was not closed.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.13	If required, BLEED off the SI Pump discharge pressure by performing the following: A. OPEN 2SJ123, TEST LINE TO CVCS HUT. B. OPEN 2SJ60, TEST LINE TO CVCS HUT. C. OPEN 2SJ158, SI HEADER COMMON DISCH. D. When 2SJ158 is open for at least 30 seconds OR PI923 and PI919 pressure decreases to 30-50 psig: 1. CLOSE 2SJ158. 2. CLOSE 2SJ60. 3. CLOSE 2SJ123.	Determines SI pump discharge pressure is lowering. Note: If pressure does NOT lower, performs the following steps: Depresses 2SJ123 TEST LINE TO CVCS HUT open PB and verifies open light illuminates and shut light extinguishes. Depresses 2SJ60, TEST LINE TO CVCS HUT open PB and verifies open light illuminates and shut light extinguishes. Depresses 2SJ158, SI HEADER COMMON DISCH open PB and verifies open light illuminates and shut light extinguishes. When 2SJ158 SI HEADER COMMON DISCH is open for at least 30 seconds: 1. Closes 2SJ158. 2. Closes 2SJ60. 3. Closes 2SJ123. Cue if required: The CRS directs you to terminate the lineup for bleeding off 21 SI pump discharge pressure.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.2.14	Direct a second Operator to PERFORM Independent Verification IAW Attachment 1, Section 2.0.	Directs a second Operator to perform Independent Verification IAW Attachment 1, Section 2.0. Cue: IV of Attachment 1 Section 2 complete.		
	5.2.15	IF TSAS 3.5.2 was entered, THEN REVIEW continued applicability of this Action Statement.	Determines TSAS 3.5.2 was not entered.		
	5.2.16	IF the Unit is in Mode 1 or 2, or Mode 3 with Pressurizer pressure >1000 psig, THEN PERFORM S2.OP-ST.SJ-0008(Q), Emergency Core Cooling - Accumulators.	Determines Unit is in Mode 1. Cue: Another NCO will perform S2.OP-ST.SJ-0008(Q), Emergency Core Cooling - Accumulators.		
	5.2.17	IF the Unit is in Mode 3 with Pressurizer pressure <1000 psig OR in modes 4, 5, 6, or Defueled, THEN PERFORM S2.OP-ST.SJ-0008(Q), Emergency Core Cooling - Accumulators, prior to exceeding 1000 psig Pressurizer pressure in Mode 3.	Determines Unit is in Mode 1.		
			Goes to Section 5.10 to raise 24 Accumulator pressure. Note: 24 Accumulator pressaure was increased to ~603 psig during fill.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	5.10.1 A/B	IF Nitrogen System is available THEN: A. ENSURE CLOSED 2NT35, N2 HDR. B. OPEN 2NT32, N2 SUPPLY.	Determines 2NT35, N2 HDR is shut by console indication. Depresses and holds open PB for 2NT32, N2 SUPPLY and verifies open light illuminates and shut light extinguishes. Note: When performing <u>next step</u> , pressure will rise very rapidly on 24 Accumulator.		
*	5.10.1 C	OPEN associated N2 Supply valve: • 21SJ93, N2 SUPPLY • 22SJ93, N2 SUPPLY • 23SJ93, N2 SUPPLY • 24SJ93, N2 SUPPLY	Depresses 24SJ93 open PB and verifies open light illuminates and shut light extinguishes.		
*	5.10.1 D	When desired Accumulator pressure is achieved, CLOSE the associated Accumulator N2 Supply valve: • 21SJ93, N2 SUPPLY • 22SJ93, N2 SUPPLY • 23SJ93, N2 SUPPLY • 24SJ93, N2 SUPPLY	Determines 24 Accumulator pressure is 625 psig on 2PI936D and 2PI937D, depresses close PB for 24SJ93, N2 SUPPLY, and verifies red open light extinguished and green shut light illuminates.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Emergency Core Cooling Systems

TASK: Raise ECCS Accumulator Level with a Safety Injection Pump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	5.10.1 E	IF required, REPEAT steps 5.10.1C and 5.10.1D to raise pressure in additional Accumulators.	Determines no other accumulators pressure will be changed.		
	5.10.1 F	CLOSE 2NT32.	Depresses close PB for 2NT32 and verifies red open light extinguished and green shut light illuminates.		
	5.10.1 G	Direct a second Operator to PERFORM Independent Verification IAW Attachment 1, Section 10.0.	Directs a second Operator to perform Independent Verification IAW Attachment 1, Section 10.0. Cue: JPM is complete.		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- SA 1. Task description and number, JPM description and number are identified.
- SA 2. Knowledge and Abilities (K/A) references are included.
- SA 3. Performance location specified. (in-plant, control room, or simulator)
- SA 4. Initial setup conditions are identified.
- SA 5. Initiating and terminating Cues are properly identified.
- SA 6. Task standards identified and verified by SME review.
- SA 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- SA 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 24 Date 3-12-13
- SA 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: *[Signature]* M.L.H.

SME/Instructor: *[Signature]* Dickman

SME/Instructor: _____

Date: 9-23-15

Date: 8/23/15

Date: _____

OPERATOR TRAINING PROGRAM JOB PERFORMANCE MEASURE

INITIAL CONDITIONS:

1. The plant is at 100% power with all systems in their normal alignment with control systems in automatic.
2. 24 Accumulator is at 58.4% level and 598.8 psig.
3. The Safety Injection System is available.
4. Nitrogen System is available.
5. RWST Boron Concentration is 2455 ppm.
6. Components Off Normal and Off Normal Tagged has been checked SAT.

INITIATING CUE:

Raise 24 ECCS Accumulator parameters to 60% level using 21 SI pump, and raise 24 Accumulator pressure to 625 psig using IAW S2.OP-SO.SJ-0002, Accumulator Operations.